Rebecca De Rosa

Curriculum Vitae

Sønderborg, Denmark
derosa.rebecca01@gmail.com
rebeccaderosa01

Summary

Cybersecurity graduate (MSc, Politecnico di Torino, 2026) working on LLM-based agents for software security. My Master's thesis built a multi-agent LLM system that augments CodeQL static analysis and detects substantially more vulnerabilities than CodeQL alone. As a research assistant at the University of Southern Denmark, I designed and evaluated an agentic vulnerability-detection pipeline with two industry partners, and I now lead the LLM development of a retrieval-augmented, human-in-the-loop document system. I am interested in how to evaluate, verify and monitor the behaviour of LLM agents, and in building the benchmarks and methods that make their failures measurable.

Education

2024–2026 **MSc in Cybersecurity**, *Politecnico di Torino*, Turin, Italy

Graduated March 2026. Master's thesis carried out at the University of Southern Denmark (SDU), Sønderborg, as an international exchange student (2025–2026).

Thesis: *A Hybrid Multi-Agent Architecture for Enhancing CodeQL Static Analysis with Large Language Models*

Supervisors: Prof. Adam Alami (SDU), Prof. Danilo Giordano and Prof. Matteo Boffa (Politecnico di Torino)

Designed a three-agent LangGraph pipeline (Analyzer, Suggestor, Creator) that validates CodeQL results, diagnoses gaps in its rules, and drafts new CodeQL queries. On 26 real-world Python files from CVEfixes (CWE-78, CWE-79, CWE-89), the Analyzer agent identified the correct vulnerability in far more files than the CodeQL baseline. Compared GPT-5.2 and GPT-4o-mini under identical settings and evaluated generated rule proposals with an LLM-as-a-judge protocol; documented over-reporting and query syntax errors as limitations.

2020–2024 **BSc in Computer Engineering**, *Politecnico di Torino*, Turin, Italy

Preprints

2026 P. M. Konrad, T. L. Adam, A. C. H. Merrild, R. Terrenzi, **R. De Rosa**, T. Tanyel, S. Ayvaz. *The Open-Box Fallacy: Why AI Deployment Needs a Calibrated Verification Regime*. arXiv:2605.10601. [arXiv]

Argues that AI authorization should rest on scoped, independently checkable verification and post-deployment monitoring rather than on mechanistic explanation alone; proposes a reportable *Verification Coverage* metric.

Research Experience

Jul 2026– **LLM Developer, Internal Project**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark

Designed the workflow and implemented a retrieval-augmented generation (RAG) pipeline that supports university employees in a document-based process. Scanned documents are first processed with OCR (PaddleOCR) and converted into structured JSON files. The extracted content is split into chunks, using a chunking method adapted to the type of text (tables or paragraphs), and the chunks are turned into semantic embeddings stored in a vector database. Each user query is embedded and matched against the database, and an LLM generates the final answer from the retrieved content. I also built an interface where users can review the answers and see the source passages highlighted in the original scanned document.

Jan–Jun 2026 **Research Assistant**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark

Co-developed, in collaboration with two software companies (Umbraco and Unik), an AI system that analyses source code to find security vulnerabilities. LLM agents detect a vulnerability, explain how it could be exploited and propose a fix, which is tested automatically before a person reviews it.

Evaluated the system on a public vulnerability benchmark and on real open-source and production code, checking its findings against known vulnerabilities and with an independent LLM judge. It uncovered a likely exploitable flaw in an open-source library, and the evaluation showed how public vulnerability labels can mislead benchmark scores.

Teaching

Jul 2026– **Teaching Assistant, Web Development**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark

Bachelor-level course. Lead the exercise sessions, working through PHP and Laravel exercises with students.

2020–2024 **Private Tutor**, Turin, Italy

Physics, Mathematics, Chemistry and Computer Science for middle and high school students.

Technical Skills

LLMs & Agents	LangChain, LangGraph, single- and multi-agent design, closed and open-source models (GPT, Claude, DeepSeek, Qwen), design and implementation of RAG-based architectures, prompt engineering
ML	PyTorch
Security	Static analysis (CodeQL), vulnerability detection
Programming	Python, Java, C, C++, PHP (Laravel); Linux

Languages

Italian	Native
English	Advanced (C1)